Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/02/2011
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of Artificial Gravity in Promoting Tissue-Regenerative Matrix-Integrin-Kinase Cell Signaling (Ground/Flight)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/01/2009	**End Date:**	11/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 ; STS-135 NOTE: End date is now 11/30/2012 per A. Chu/ARC (Ed., 4/11/12) NOTE: Start/end dates changed to 6/1/09-5/31/12 from 5/1/09-4/30/12, per A. Chu/ARC (6/2009)		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Globus, Ruth (NASA Ames Research Center)		
Grant/Contract No.:			
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Performance Goal Text:			

Task Description:	Our broad working hypothesis underlying this proposal is that in space, the mechanical unloading caused by microgravity can affect the growth and regenerative health of a broad range of mammalian tissues. Conversely, we also hypothesize that gravity on earth activates a key, complex cell-signaling pathway responsible for maintaining normal regenerative tissue growth, the extracellular matrix(ECM)-integrin-kinase signaling pathway. The work we propose here is centered on the specific hypothesis that somatic stem cells responsible for tissue regeneration may require the mechanical stimulation of gravity to proliferate and differentiate, and to regenerate tissues at normal rates, and that microgravity may slow down tissue regenerative processes. Our preliminary results from spaceflight experiments with amphibian (newt) tail regeneration during the Foton M2 and Foton M3 missions, suggests that spaceflight significantly retards tissue regeneration. These findings strongly suggest that normal long-term homeostatic tissue regenerative processes may be sensitive to the absence of gravity-generated forces in space, and that this may cause tissue degeneration. In this proposal we seek to gather further evidence about the role of gravity-generated forces in promoting tissue regeneration, and to investigate the molecular mechanisms by which gravity in stem cells is translated into proliferation and differentiation. Specifically we proposed: AIM 1. To test the hypothesis that gravity-dependent mechanical loading stimulates cell proliferation and differentiation in the newt tail regeneration spaceflight model via the matrix-integrin-kinase signaling pathway AIM 2. To test the hypothesis that gravity-dependent mechanical loading activates the matrix-integrin-kinase signaling pathway in pluripotent mouse and newt stem cells and increases their proliferation and differentiation. The potential benefit of concluding these studies successfully is to define the contribution of the ECM-integrin-kinase pathway in somatic stem cells as a link between gravity sensing and tissue regeneration, which would help define risk for long-term tissue degeneration in space and provide a rationale to develop novel countermeasures. NOTE: This proposal has a companion spaceflight component entitled, "The Effects of Microgravity on the Tissue Regenerative Potential of Stem Cells--Flight." Having completed the initial data analysis of the STL-SCR flight experiments conducted on STS-131, and for context, we also report here on the companion flight proposal progress and results. NOTE that the Flight and Ground portions of the project were combined in April 2011 for reporting purposes (Ed. 7/11/2011).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We believe the studies underway in this project are highly relevant to NASA because the hypothesis we are testing test, if validated, will be of key importance to understanding and maintenance of human tissue regenerative homeostasis and health during long-term spaceflight. The data we have already obtained in complementary spaceflight and ground studies clearly and reproducibly show that tissue regeneration in animal and cell models is significantly slower in space, and that gravity-generated forces are of key importance to stimulate the growth, proliferation, and differentiation of cells and tissues. In this study we asked the specific question of whether tissue regeneration in vivo, in the newt tail regeneration model, and regenerative somatic stem cell models, in vitro respond to hypergravity with increased regenerative proliferation, and if so by what molecular signaling mechanisms. These questions are of key importance because hypergravity in space is thought to be one of the few methods capable of compensating for the degenerative tissue conditions now know to be induced by long-term spaceflight. The results obtained in this grant thus far, with stem cell and newt tail regeneration experiments under hypergravity conditions, strongly support our initial hypothesis. The in-vivo newt tail regeneration in particular, for the first time show vertebrate tissue regeneration being enhanced by gravity stimulation. In addition, companion spaceflight experiments in the STS-131 STL-SCR, studying stem cell growth and differentiation, also point to microgravity as causing a failure of differentiation of tissue precursor cells. In conclusion, this study, already shows promise of having great impact in this field by providing specific answers to the important question of whether and how gravity can promote regenerative proliferation and differentiation in relevant cell and animal models of regeneration. These novel insights into the potentially broad role of adult stem cell health in spaceflight-induced tissue degenerative conditions are likely to be an important factor in shaping the direction of future approaches to health during spaceflight, as well as being translated into medical advances in tissue regenerative health.
Task Progress:	Because of launch schedule for STS 131, the initial period of the 2010-2011 grant year from April to June 2010 was dedicated to conducting the spaceflight companion portion to this grant, specifically the Space Tissue Loss, Stem Cell Regeneration experiment on the STS-131 mission. Specifically we used the Cell Culture Module (CCM) hardware to differentiate mouse embryonic stem cells in space, using an experimental model based on the formation of embryoid bodies, or spheres of various differentiated tissue lineages and cell types. The experimental method was initially developed for our ground studies and adapted for flight. The flight experiment was successfully conducted and we recovered both bioreactors with cells fixed during spaceflight, with RNA Later, for gene expression analysis, as well as live cells for cell biology assays conducted post-flight. In brief we found that on ground controls a variety of genes markers of tissue differentiation quantified using real time qPCR appeared normally as embryoid bodies differentiated. These markers included specific genes for endoderm, mesoderm, ectoderm, and specific cell types within these tissues. In addition in controls we observed the disappearance of stem cell markers. In spaceflight samples, however, nearly of 80% of the tissue lineage markers characteristic of embryoid body differentiation did not appear, and stem cell markers remained elevated, suggesting spaceflight inhibited normal tissue differentiation and maintained "stemmness" of the differentiating cultures. Measurements of cell number, DNA content, and glucose consumption also revealed r a decrease in cell proliferation in space, but no changes in cell viability or apoptosis. Finally, live cells recovered post-flight from embryoid bodies in spaceflight bioreactors, when cultured showed greater (double) potential for differentiation of contractile cardiomyocyte colonies. These results strongly support our hypothesis that the tissue regenerative potential of stem cells may be compromised during spaceflight. Following the STS-131 experiments, in the summer of 2010, we focused our efforts on hypergravity experiments using the same embryonic stem cell experimental model. We conducted six short-term experiment under hypergravity and showed a 20% to 40% increase in stem cell proliferation in response to hypergravity (10g). Additional longer-term embryoid body experiments for gene expression analysis in hypergravity are now underway. Late in 2010/early 2011 we developed a new experimental model for stem cell differentiation into a single keratinocyte lineage,

	which is more suitable to long-term hypergravity experiments than embryoid bodies, and lends itself more readily to the molecular dissection of signaling pathways. After successful testing, this model was recently proposed in a proposal selected for a spaceflight experiment " re-fly" of the STL-SCR experiment now on the upcoming STS-135 mission. These experiments are currently undergoing pre-flight verification. In addition to the stem cell biology component of this proposal, we also made major progress in the whole animal stem cell-based tissue regeneration experiments, using newt tail re-growth under hypergravity conditions. Specifically we used the NASA Ames 8ft centrifuge facility to centrifuge at 2g, 16 newts with tail tips surgically removed for a period of 12 days, under conditions nearly identical to the Foton M3 spaceflight experiment, in which we observed a deficit in tail regeneration during microgravity. This experiment was successfully conducted in October 2010, and we already analyzed tail volumes by soft tissue microCT and are currently conducting bone microCT, BrdU incorporation analysis, and bone sectioning for histology. The initial results from soft tissue microCT show tail volumes in hypergravity animals increased significantly, supporting our hypothesis that gravity-generated forces promote stem cell-based tissue growth and regeneration. Analysis of the newt experiment will continue through the fall of 2011.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Finkelstein H, Dvorochkin N, Yousuf R, Globus RK, Almeida EA. "Spaceflight Reduces the Tissue Regenerative Potential of Stem Cells by Decreasing Proliferation and Increasing Early Differentiation." Stem Cells Biology Poster Session. 50th Annual Meeting of the American Society for Cell Biology, Philadelphia, PA, December 11-15, 2010. Molecular Biology of the Cell 2010 Dec 15;21(24):2494. http://dx.doi.org/10.1091/mbc.E10-10-0839 , Dec-2010
Abstracts for Journals and Proceedings	Finkelstein H, Dvorochkin N, Yousuf R, Globus RK, Almeida EA. "Spaceflight Reduces Tissue Regenerative Potential of Stem Cells." Poster Session, 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010. Program and abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, 2010. , Nov-2010